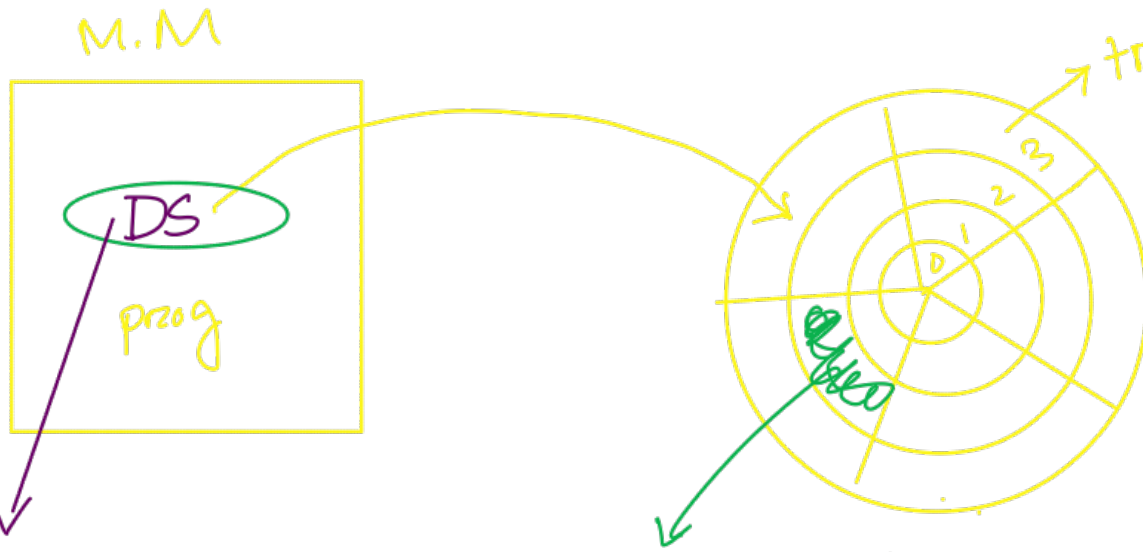
Spinning disk → change sectores

moving head → change track

DBMS → organizing data efficiently on disk



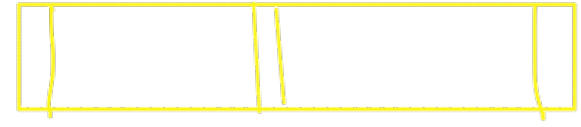
Data inside M.M directly used by prog.

block

address:

sector #, track #

pizza sect



512 byte

offset

Employee

eid — 10

name — 50



dept – 10

sec – 8

addrs – 50

---

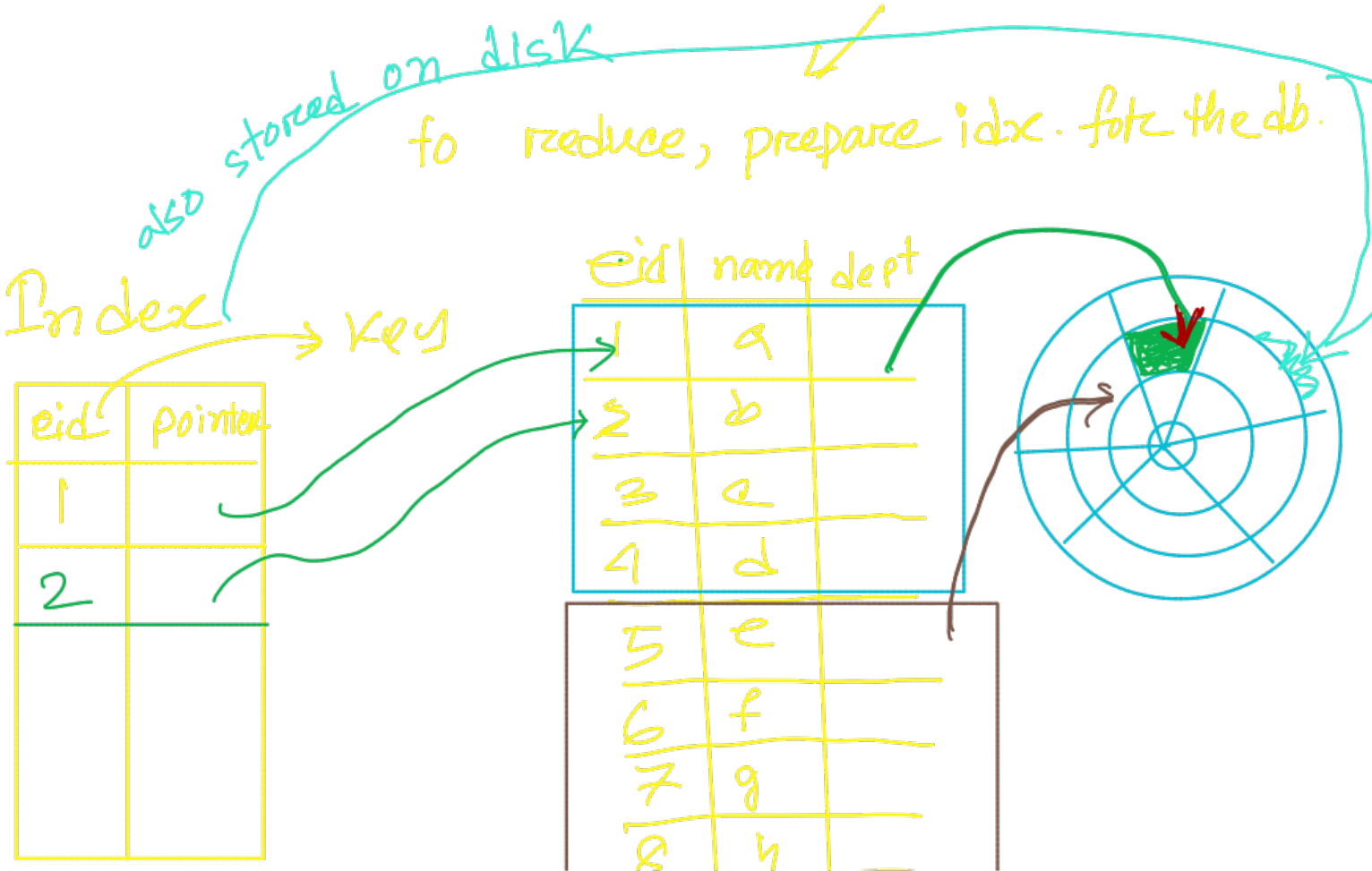
size – 128 bytes

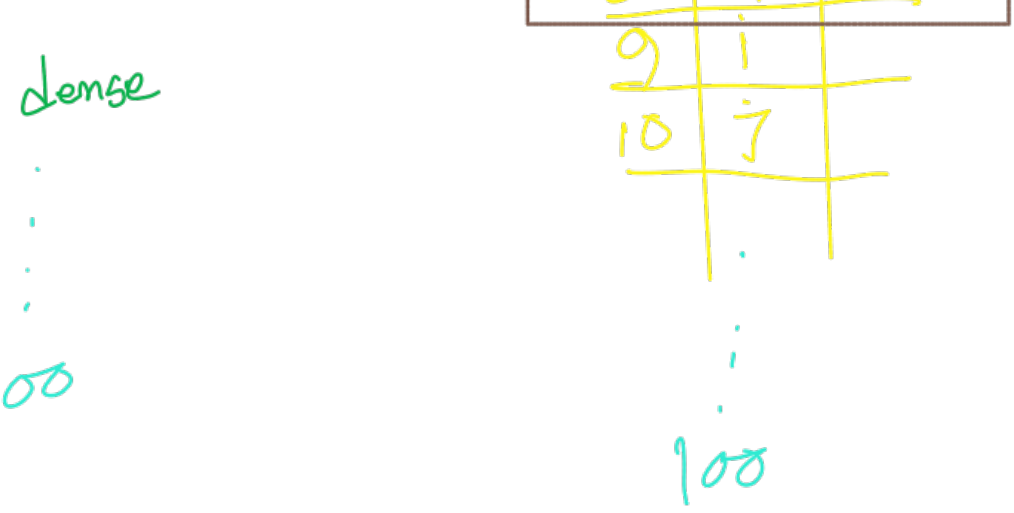
of records/Block = $\frac{512}{128}$

= 4

For 100 records → $\frac{100}{4}$ = 25 blocks.

Select From employee → search all 25 blocks.





eid → 10

rec ptr → 6

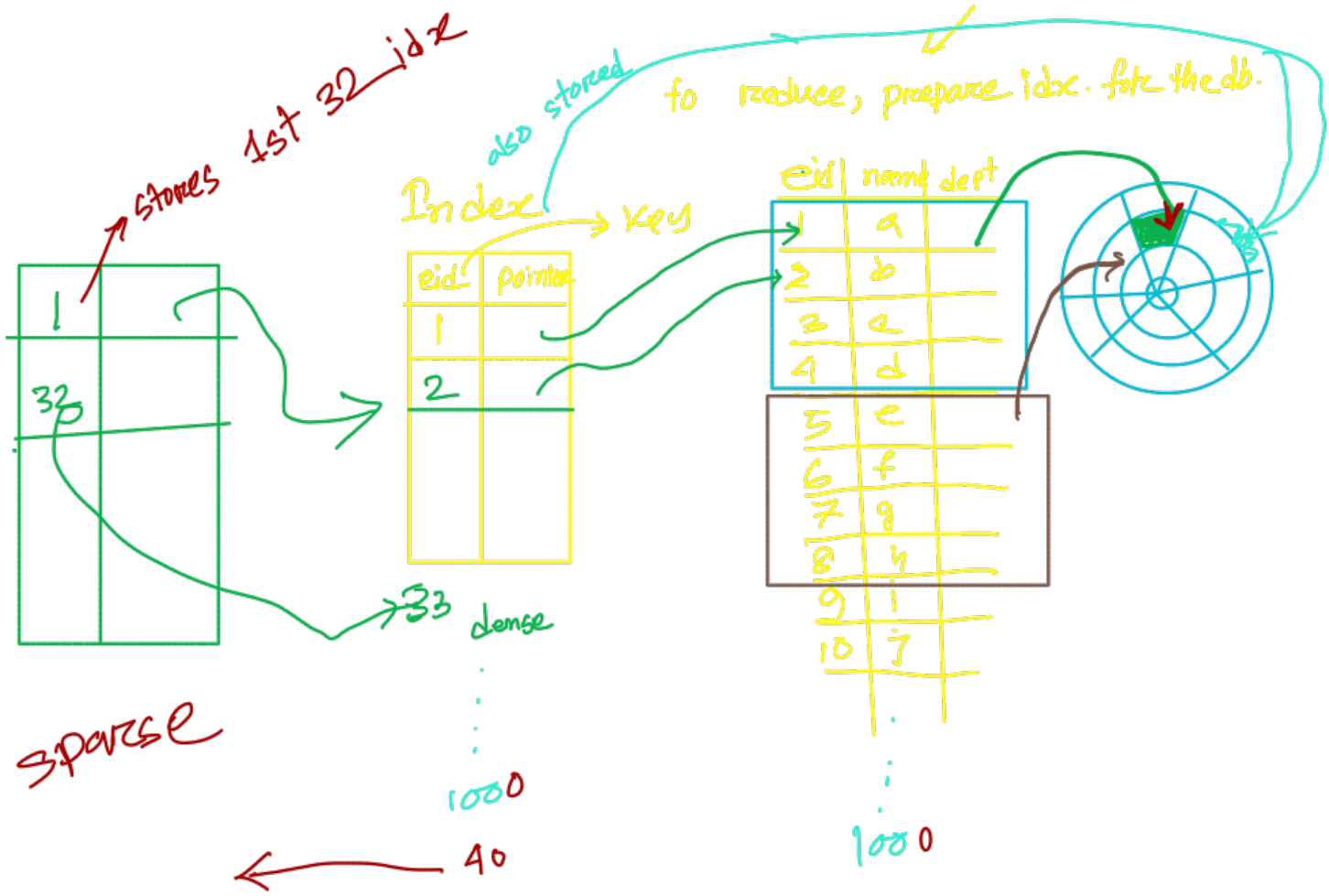
16 bytes

no. of entries/block = $\frac{512}{16}$ = 32 entries in a block

For 100 entries → $\frac{100}{32} = 3.2$ = 3+ blocks = 4 blocks of idx.

So search A block then the block the certain ptr is pointing to → total 5 blocks

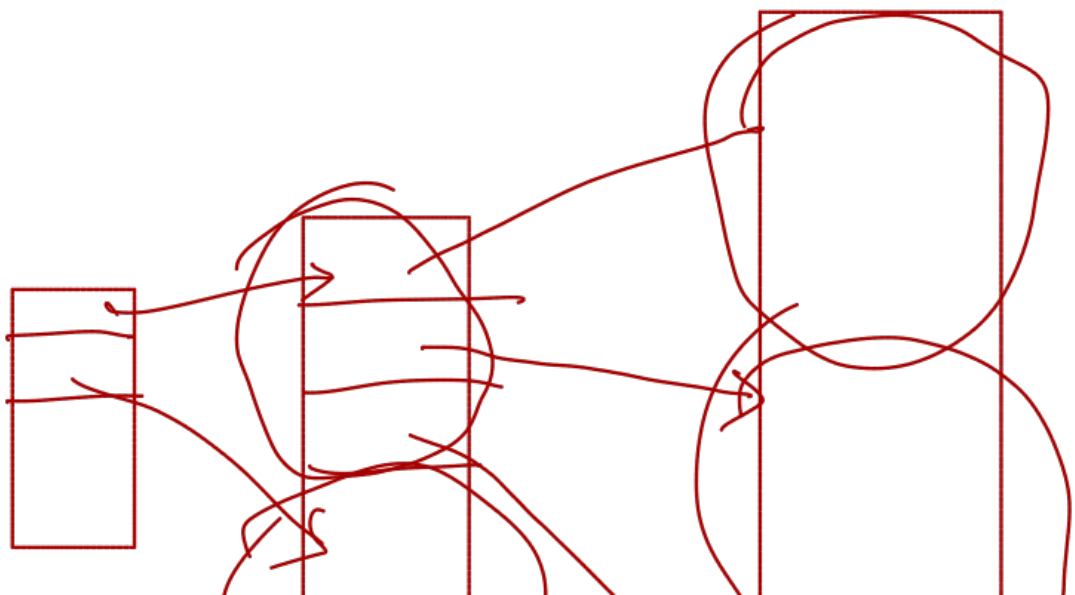
Multilevel idx

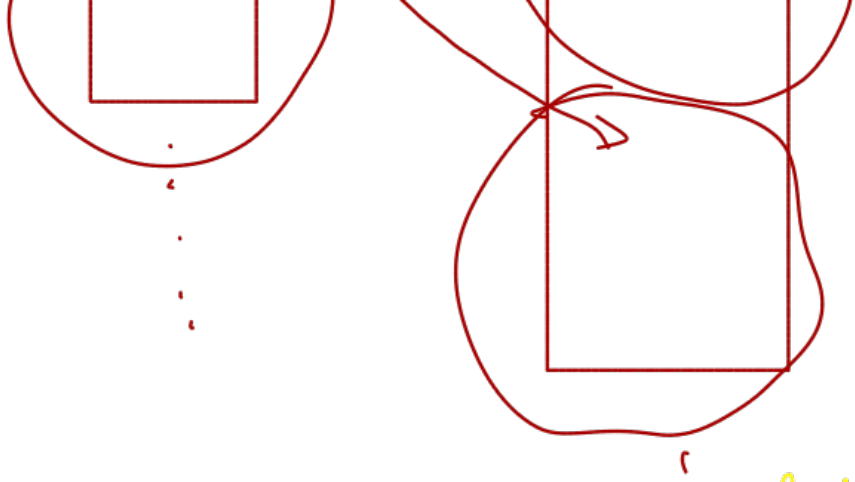


$$\frac{40}{32} = 1 + \text{block}$$

$$= 2 \text{ block}$$

So $2 + 1 + 1 = 4$





we want high level added & deleted automatically. (Self managed multilevel idx

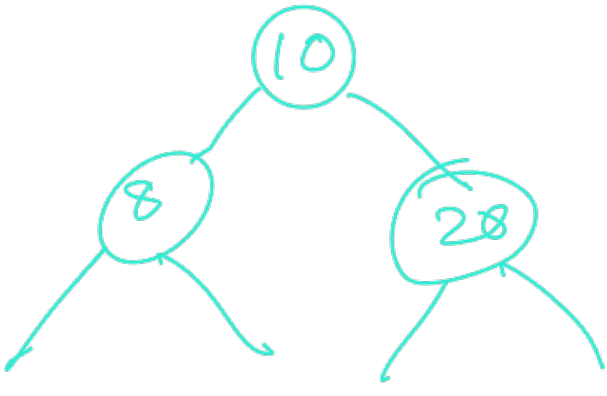
M way search tree

m children (max)

(m-1) Key

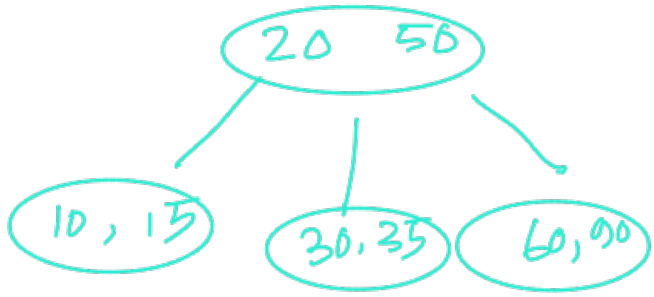
(m-1) rp

BST



1 key per node

2 children "

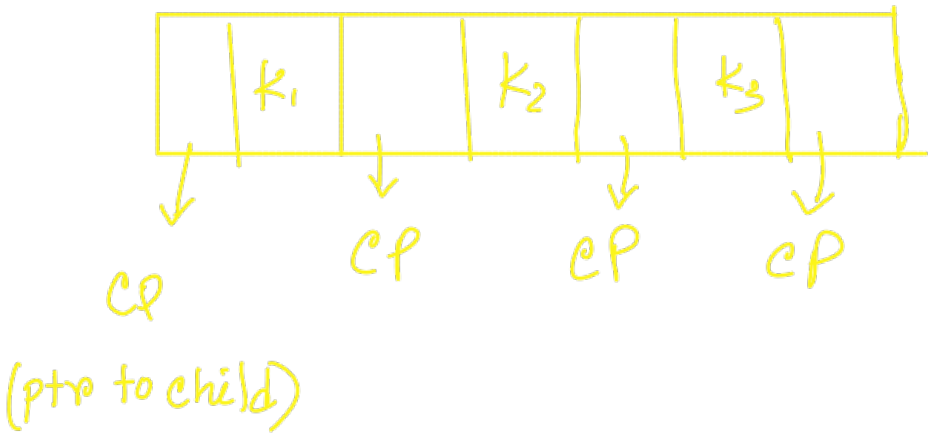


2key

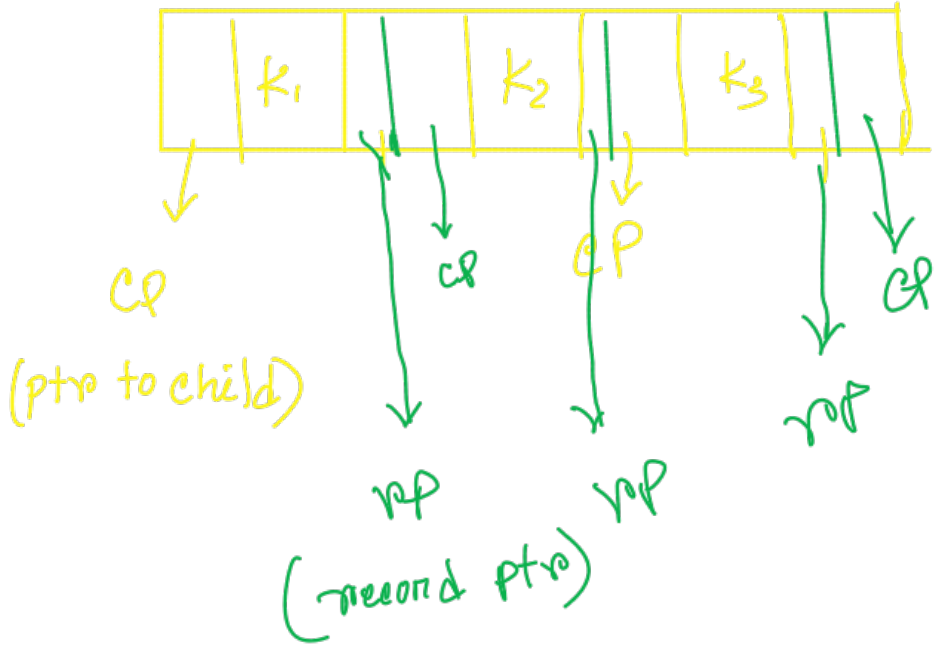
3 children

3 way search tree

4 way ST

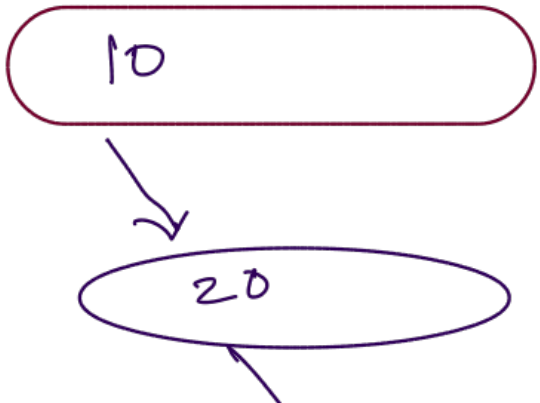


for idx :



10 way ST

Keys; 10, 20, 30



no control how to create (write)

30

Rules + guidelines ⟹ Btrees

1. $\lceil m/2 \rceil$ → ceil children must [if current node is half filled with children then only go to next node

2. root can have min 2 children, 1 key ✓

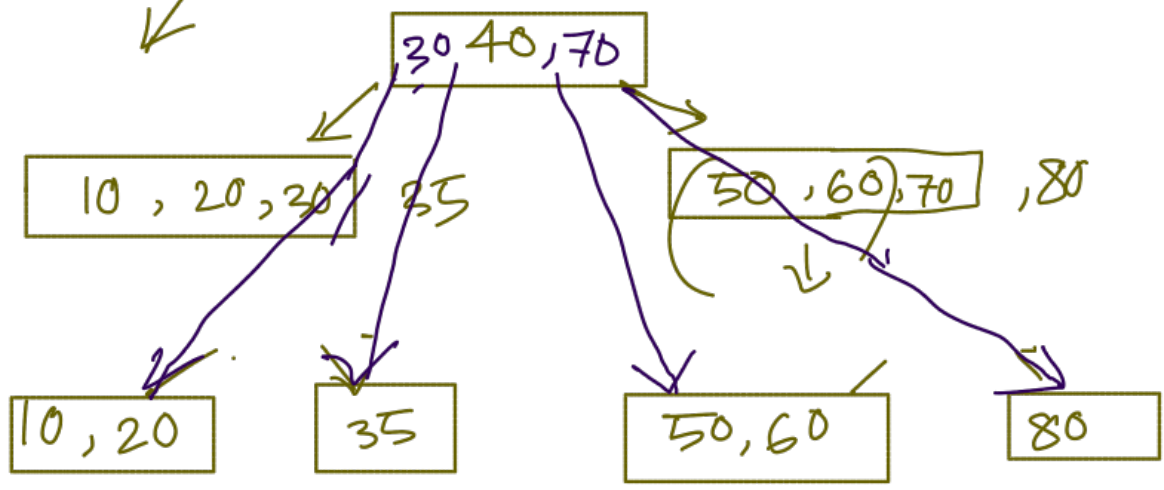
3. All leaves at same level.

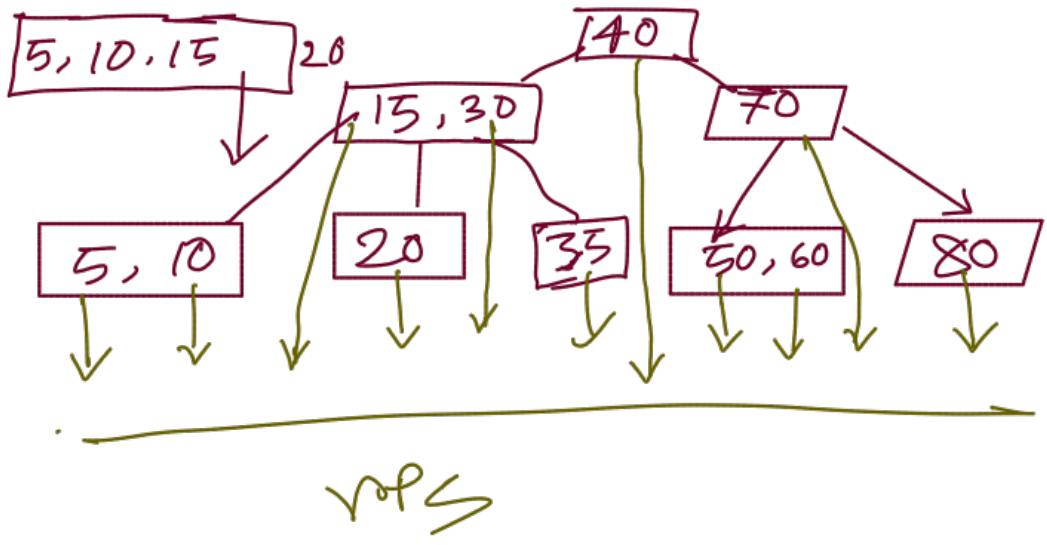
4. Bottom up creation.

$m = 4$

Key: 10, 20, 40, 50, 60, 70, 30, 35



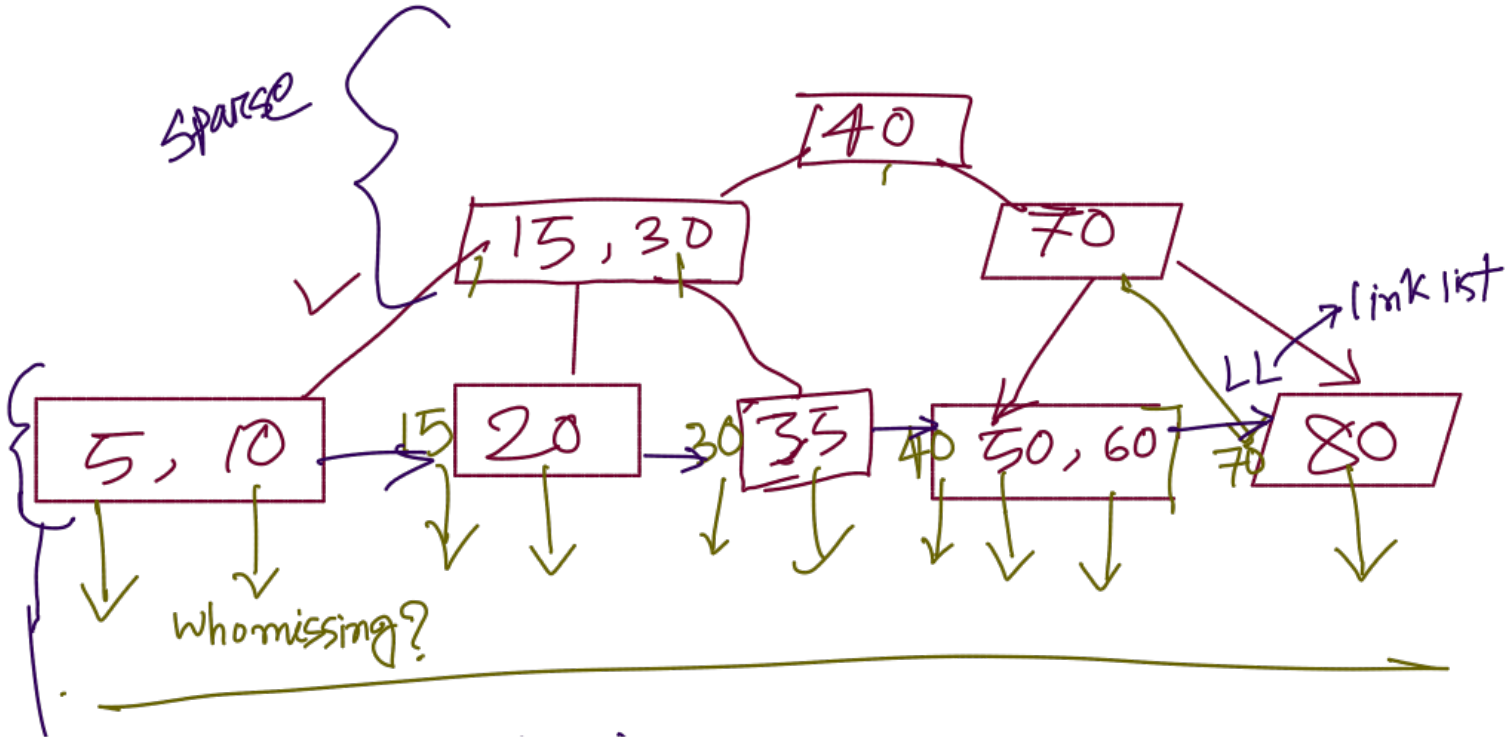




vvs

B+ tree: Not rec ptrs from every nodes.

only from leaf nodes



↳ dense (multilevel idx)

every key will have its copy in the leaf nodes.

Q) $t_1 \rightarrow 9, 18, 42, 51$

$t_2 \rightarrow 65, 70, 81$

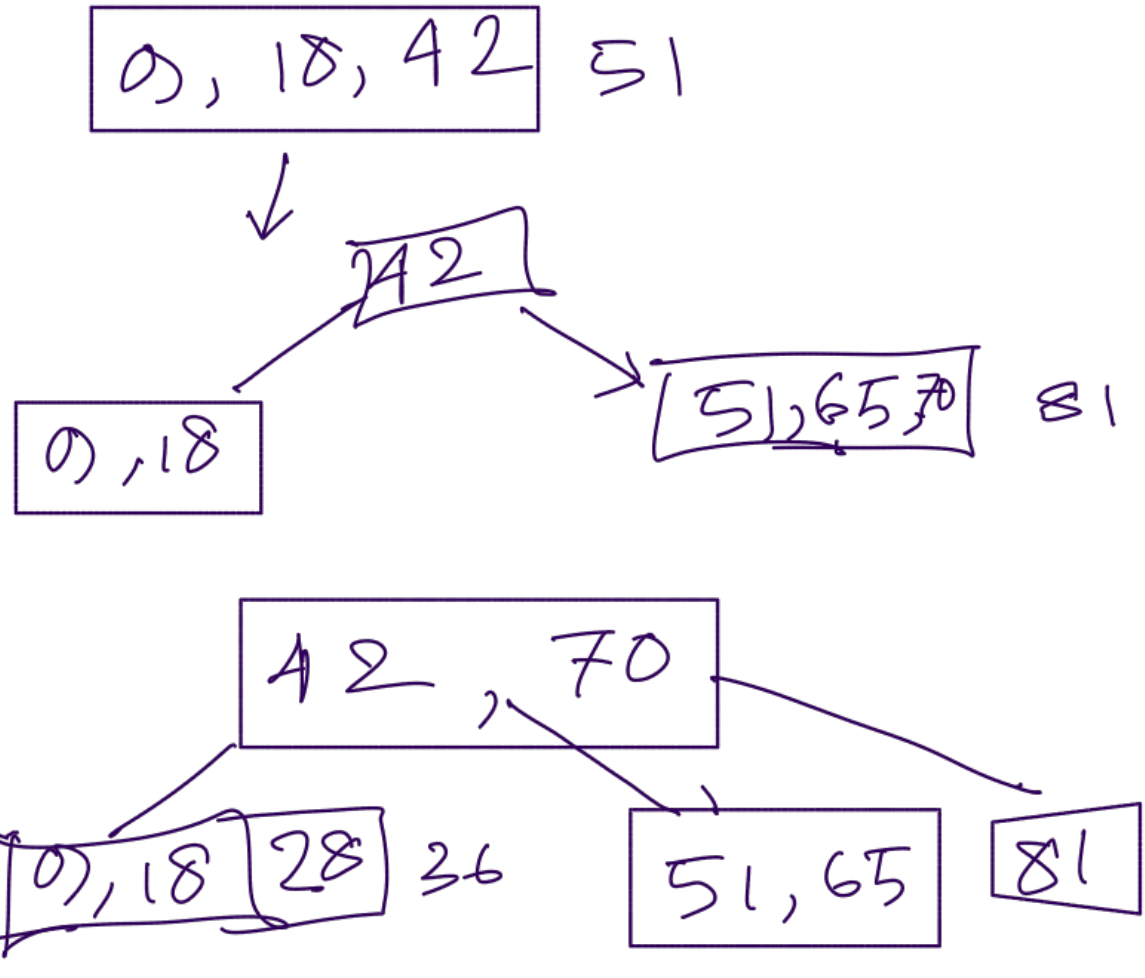
$t_3 \rightarrow 28, 36$

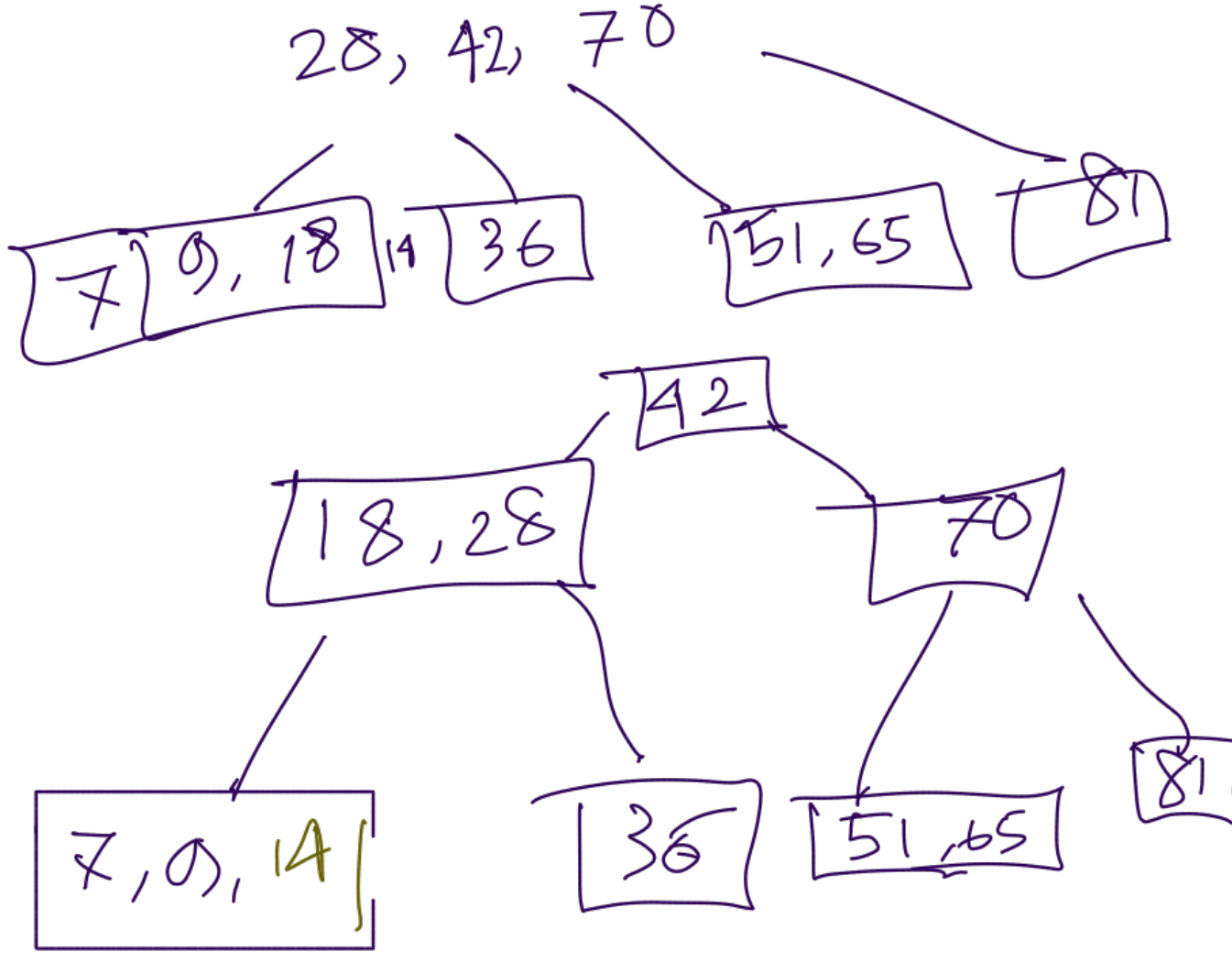
$t_4 \rightarrow 7, 14$

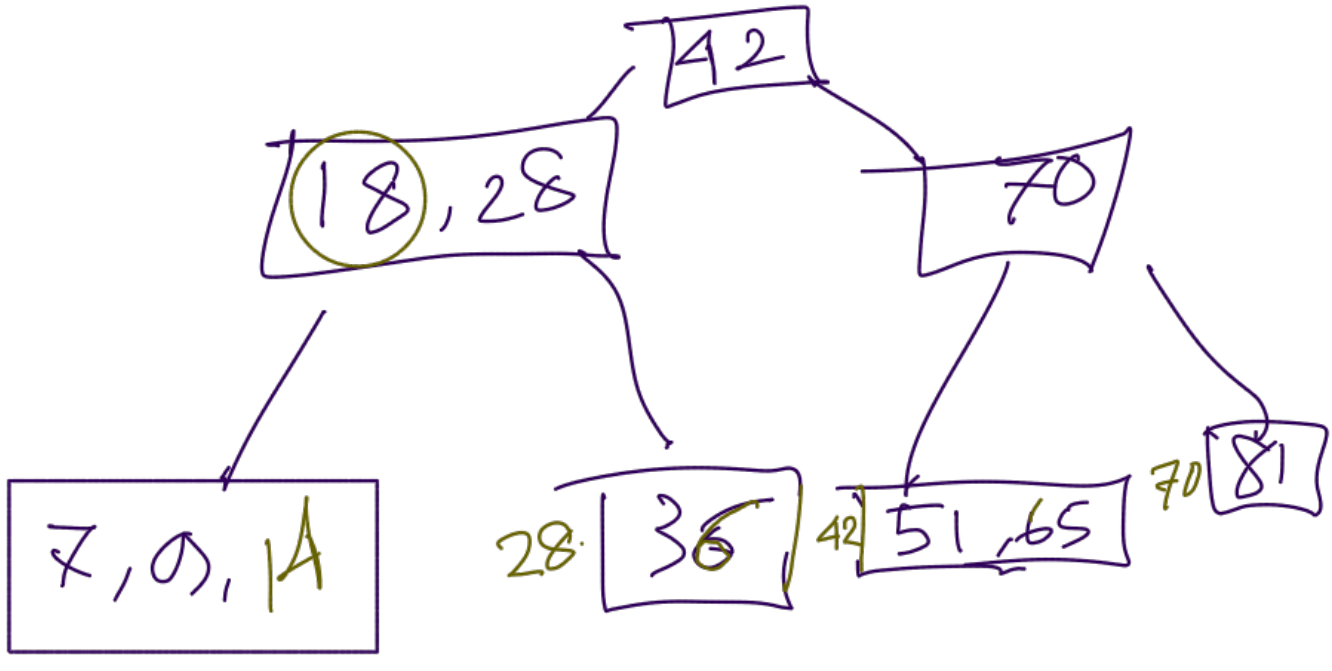
$m = 4$

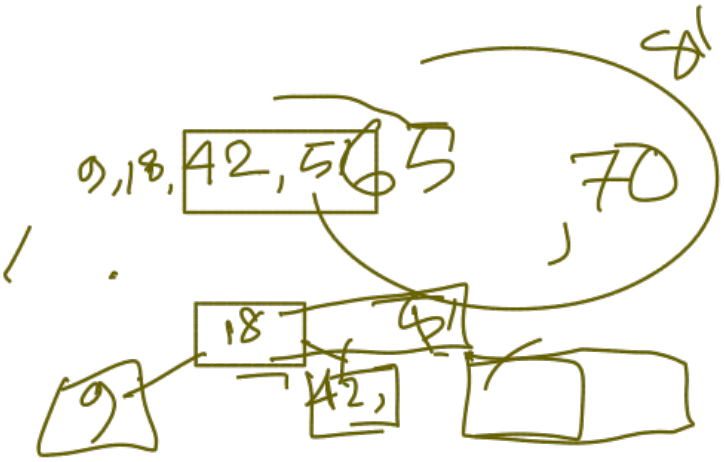
3 key

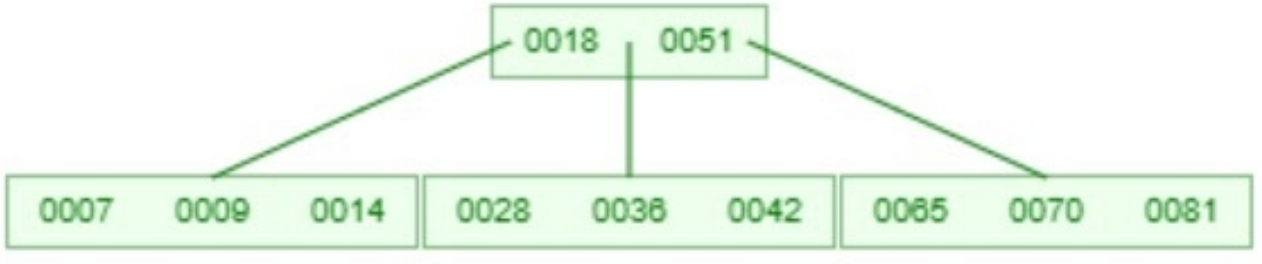
4 children

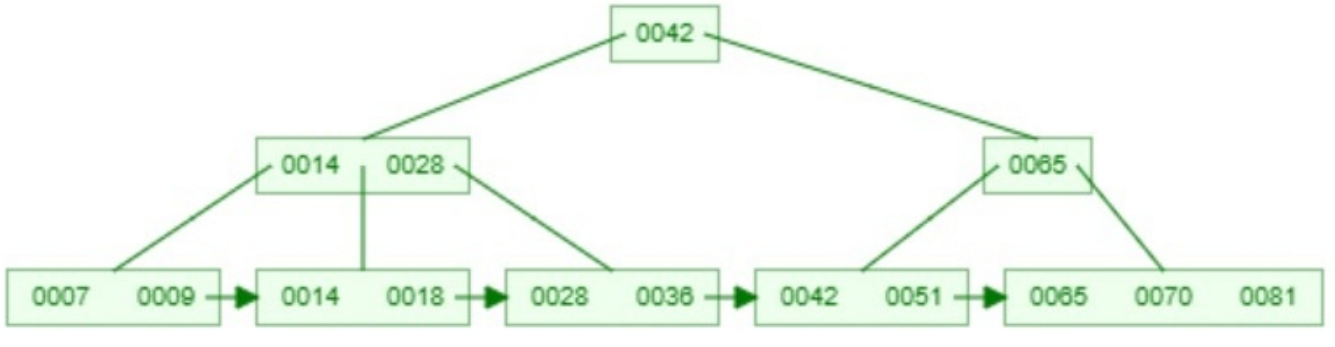


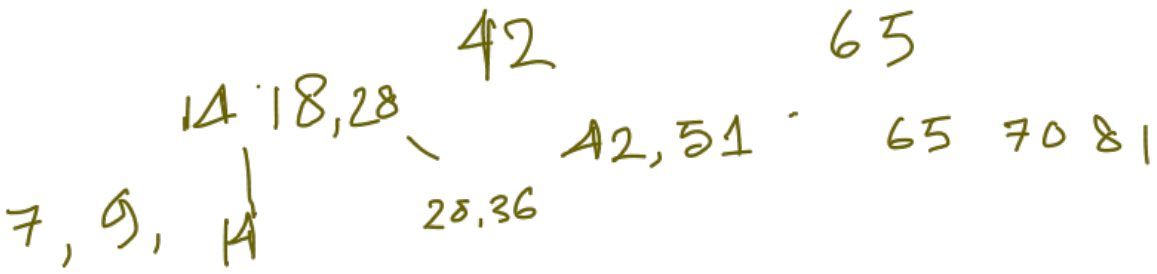

28, 42, 70
7
9, 18
14
36
51, 65
81
42
18, 28
70
7, 0, 14
36
51, 65
81

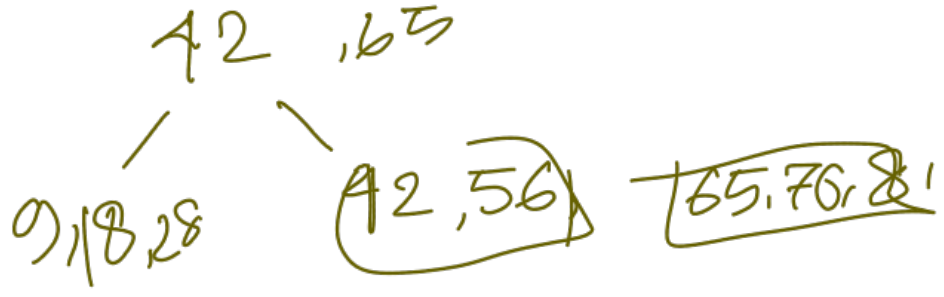

42
18, 28
70
7, 0, 14
28
36
42
51, 65
70
81

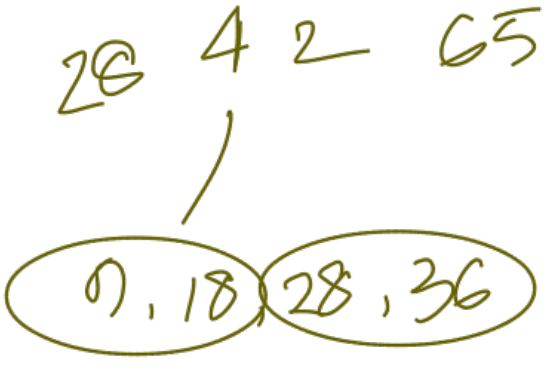



0018
0051
0007
0009
0014
0028
0036
0042
0065
0070
0081


0042
0014
0028
0065
0007
0009
0014
0018
0028
0036
0042
0051
0065
0070
0081


42
65
14 18,28
42,51
65 70 81
7, 9, 14
28,36


42 ,65
9,18,28
42,56
65,76,81


26 42 65
9, 18
28, 36